

Problem 2

Problem 2

Problem 1 Evaluate the following definite integrals:

(a) $\int_{-2}^1 t^2 \sin(t^3) dt$

Explanation. Set $v = v(t) = t^3$, where “v(t)” is meant to indicate that the variable v is really a function of t . Then $dv = 3t^2 dt$. But notice that

$$\begin{aligned} v(-2) &= (-2)^3 = -8 \\ v(1) &= (1)^3 = 1. \end{aligned}$$

Then

$$\begin{aligned} \int_{-2}^1 t^2 \sin(t^3) dt &= \frac{1}{3} \int_{-2}^1 3t^2 \sin(t^3) dt \\ &= \frac{1}{3} \int_{-8}^1 \sin(v) dv \\ &= -\frac{1}{3} \left[\cos(v) \right]_{-8}^1 \\ &= -\frac{1}{3} (\cos(1) - \cos(-8)) \end{aligned}$$

(b) $\int_0^{1/2} \frac{13e^u}{3e^u - 5} du$

Explanation. Let $w = 3e^u - 5$. Then

$$\begin{aligned} dw &= 3e^u du \\ w(0) &= 3e^0 - 5 = 3 - 5 = -2 \\ w\left(\frac{1}{2}\right) &= 3e^{\frac{1}{2}} - 5. \end{aligned}$$

Substituting this into the original integral gives

$$\begin{aligned}
 \int_0^{1/2} \frac{13e^u}{3e^u - 5} du &= \frac{13}{3} \int_0^{1/2} \frac{3e^u}{3e^u - 5} du \\
 &= \frac{13}{3} \int_{-2}^{3e^{1/2} - 5} \frac{1}{w} dw \\
 &= \frac{13}{3} \left[\ln |w| \right]_{-2}^{3e^{1/2} - 5} \\
 &= \frac{13}{3} \left(\ln |3e^{1/2} - 5| - \ln |-2| \right) \\
 &= \frac{13}{3} \left(\ln \left(5 - 3e^{1/2} \right) - \ln(2) \right).
 \end{aligned}$$

NOTE: $3e^u - 5$ is always negative on our interval $\left[0, \frac{1}{2}\right]$. Therefore, the denominator is never 0 and the integrand (function) is continuous on $\left[0, \frac{1}{2}\right]$.

(c) $\int_0^{\pi/2} \frac{\cos(x)}{4 + \sin(x)} dx$

Explanation. Let $w = 4 + \sin(x)$, so that $dw = \cos(x)dx$. When $x = 0$, $w = 4$ and when $x = \frac{\pi}{2}$, $w = 5$.

$$\begin{aligned}
 \int_0^{\pi/2} \frac{\cos(x)}{4 + \sin(x)} dx &= \int_4^5 \frac{dw}{w} \\
 &= \left[\ln |w| \right]_4^5 \\
 &= \ln(5) - \ln(4) \\
 &= \ln \left(\frac{5}{4} \right).
 \end{aligned}$$

(d) $\int_1^4 \frac{e^{\sqrt{x}}}{3\sqrt{x}} dx$

Explanation. Set $u = \sqrt{x}$. Then

$$\begin{aligned}
 du &= \frac{1}{2\sqrt{x}} dx \\
 u(1) &= \sqrt{1} = 1 \\
 u(4) &= \sqrt{4} = 2.
 \end{aligned}$$

Substituting these into the original integral gives:

$$\begin{aligned}
 \int_1^4 \frac{e^{\sqrt{x}}}{3\sqrt{x}} dx &= \frac{2}{3} \int_1^4 \frac{e^{\sqrt{x}}}{2\sqrt{x}} dx \\
 &= \frac{2}{3} \int_1^2 e^u du \\
 &= \frac{2}{3} \left[e^u \right]_1^2 \\
 &= \frac{2}{3} (e^2 - e)
 \end{aligned}$$

$$(e) \int_{\pi/3}^{\pi/2} \sin(x) \sec^2(\cos(x)) dx$$

Explanation. Let $v = \cos(x)$. Then

$$\begin{aligned}
 dv &= -\sin(x)dx \\
 v\left(\frac{\pi}{3}\right) &= \frac{1}{2} \\
 v\left(\frac{\pi}{2}\right) &= 0.
 \end{aligned}$$

So

$$\begin{aligned}
 \int_{\pi/3}^{\pi/2} \sin(x) \sec^2(\cos(x)) dx &= - \int_{1/2}^0 \sec^2(v) dv \\
 &= - \left[\tan(v) \right]_{1/2}^0 \\
 &= - \left(0 - \tan\left(\frac{1}{2}\right) \right) \\
 &= \tan\left(\frac{1}{2}\right).
 \end{aligned}$$